```c
#include <stdio.h>
#include <stdlib.h>

int main() {
    int *ptr = malloc(5 * sizeof(int));

    if (ptr == NULL) {
        perror("malloc");
        return 1;
    }

    for (int i = 0; i < 5; i++)
        ptr[i] = i + 1;

    printf("Values: ");
    for (int i = 0; i < 5; i++)
        printf("%d ", ptr[i]);

    printf("\n");

    free(ptr);
    ptr = NULL;

    printf("Memory released successfully.\n");
    return 0;
}
```

```
#!/bin/bash

gcc -Wall -Wextra q22a.c -o q22a || exit 1

./q22a > output.txt

if grep -q "Memory released successfully." output.txt; then
    echo "Test 1: PASS"
else
    echo "Test 1: FAIL"
fi

if grep -q "1 2 3 4 5" output.txt; then
    echo "Test 2: PASS"
else
    echo "Test 2: FAIL"
fi
```

```
sravya@Sravya1955:~/os$ nano q22a.c
sravya@Sravya1955:~/os$ gcc -g q22a.c -o q22a
sravya@Sravya1955:~/os$ ./q22a
Values: 1 2 3 4 5
Memory released successfully.
sravya@Sravya1955:~/os$ valgrind --leak-check=full ./q22a
==2565== Memcheck, a memory error detector
==2565== Copyright (C) 2002-2022, and GNU GPL'd, by Julian Seward et al.
==2565== Using Valgrind-3.22.0 and LibVEX; rerun with -h for copyright info
==2565== Command: ./q22a
==2565==
Values: 1 2 3 4 5
Memory released successfully.
==2565==
==2565== HEAP SUMMARY:
==2565==     in use at exit: 0 bytes in 0 blocks
==2565==   total heap usage: 2 allocs, 2 frees, 1,044 bytes allocated
==2565==
==2565== All heap blocks were freed -- no leaks are possible
==2565==
==2565== For lists of detected and suppressed errors, rerun with: -s
==2565== ERROR SUMMARY: 0 errors from 0 contexts (suppressed: 0 from 0)
sravya@Sravya1955:~/os$ nano test_q22.sh
sravya@Sravya1955:~/os$ chmod +x test_q22.sh
sravya@Sravya1955:~/os$ ./test_q22.sh
Test 1: PASS
Test 2: PASS
sravya@Sravya1955:~/os$
```

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